

OCR Further Mathematics
Pure Core - Further Algebra
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
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Question			Answer/Indicative content	Marks	Part marks and guidance	
1	a		e.g. $\alpha^2 + \beta^2 + \gamma^2 = -5$ means that at least one root is complex But complex roots come in complex pairs so there are 2 complex roots. Given that there are 3 roots and 2 are complex one is real.	B1 (AO 2.4) B1 (AO 2.4) B1 (AO 2.4) [3]		
	b		$\alpha + \beta + \gamma = 3$ $(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \beta\gamma + \gamma\alpha)$ $9 = -5 + 2(\alpha\beta + \beta\gamma + \gamma\alpha)$ But $k = \alpha\beta + \beta\gamma + \gamma\alpha$ $\Rightarrow k = 7$	B1 (AO 1.1) M1 (AO 3.1 a) A1 (AO 1.1) [3]	Attempt to obtain identity and substitute Condone missing 2 and sign errors	
	c		$\left(\frac{1}{u}\right)^3 - 3\left(\frac{1}{u}\right)^2 + 7\left(\frac{1}{u}\right) - 5 = 0$ $\Rightarrow -5u^3 + 7u^2 - 3u + 1 = 0$ oe Alternate method $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{7}{-5}$ $\frac{1}{\alpha\beta\gamma} = \frac{1}{-5}$ Answer as above	M1 (AO 1.1) A1 (AO 1.1) Alternate method M1 A1 [2]	For the substitution “=0” not necessary here but needed for A1 Allow in terms of z Allow ft from <i>their</i> k in part above Alternate method For calculating the sum, product and sum of product of pairs of reciprocals of α , β , γ	
			Total	8		



Question		Answer/Indicative content	Marks	Part marks and guidance		
2	a	DR $\frac{x^3 + x^2 + 9x - 1}{x^3 + x^2 + 4x + 4} \equiv \frac{x^3 + x^2 + 4x + 4 + 5x - 5}{x^3 + x^2 + 4x + 4}$ $= 1 + \frac{5x - 5}{x^3 + x^2 + 4x + 4}$ So $A = 1$, $B = 5$ and $C = -5$	B1 (AO3.1a) [1]	Attempt to divide out improper fraction. Could be by symbolic division or other valid method (eg comparing coefficients or substitution of values for x)	Allow embedded answers	
	b	DR $x^3 + x^2 + 4x + 4 = (x+1)(x^2 + 4)$ $\frac{5x - 5}{x^3 + x^2 + 4x + 4} = \frac{D}{x+1} + \frac{Ex + F}{x^2 + 4}$ $D(x^2 + 4) + (x+1)(Ex + F) = 5x - 5$ $x = -1 \Rightarrow 5D = -10 \Rightarrow D = -2$ $x = 0 \Rightarrow -2 - F = -5 \Rightarrow F = -3$ $x^2: D + E = 0 \Rightarrow E = 2$ $1 - \frac{2}{x+1} + \frac{2x+3}{x^2+4}$	B1 (AO3.1a) M1 (AO1.2) A1 (AO1.1) A1 (AO1.1) A1 (AO1.1) [5]	Correct factorisation of cubic seen in working Correct form for partial fractions equated to their remainder rational fraction from (a). Follow through their division and factorisation. Or equivalent to find D correctly.	Could be from improper fraction Allow ft A1 for second and third coefficients found. Or $1 - \frac{2}{x+1} + \frac{2x}{x^2+4} - \frac{3}{x^2+4}$	



Question		Answer/Indicative content	Marks	Part marks and guidance	
	c	DR $\int_0^2 \frac{x^3 + x^2 + 9x - 1}{x^3 + x^2 + 4x + 4} dx = \int_0^2 1 - \frac{2}{x+1} + \frac{2x}{x^2+4} + \frac{3}{x^2+4} dx$ $= \left[x - 2 \ln(x+1) + \ln(x^2+4) + \frac{3}{2} \tan^{-1}\left(\frac{x}{2}\right) \right]_0^2$ $\left(2 - 2 \ln 3 + \ln 8 + \frac{3\pi}{8} \right) - \ln 4$ $2 + \ln\left(\frac{2}{9}\right) + \frac{3}{8}\pi$	*M1 (AO3.1 a) dep*M1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [4]	Split term with $x^2 + 4$ in denominator and $ax + b$ in numerator Correctly integrate <i>their</i> expression (ignore limits) Correctly substitute limits to produce exact values and evaluate <i>their</i> $\tan^{-1}$ term $a = 2, b = \frac{2}{9}, c = \frac{3}{8}$	
		Total	10		

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Question		Answer/Indicative content	Marks	Part marks and guidance		
3	a	DR $(r+2)(r-1)$ $\frac{A}{r-1} + \frac{B}{r+2}$ $A=1, B=1$ $\therefore \sum_{r=5}^n \frac{3}{r^2+r-2} = \sum_{r=5}^n \frac{1}{r-1} - \sum_{r=5}^n \frac{1}{r+2}$ $= \sum_{r=5}^n \frac{1}{r-1} - \sum_{r=8}^{n+3} \frac{1}{r-1}$ $= \sum_{r=5}^7 \frac{1}{r-1} - \sum_{r=n+1}^{n+3} \frac{1}{r-1}$	B1 (AO1.1) M1 (AO1.1) A1 (AO1.1) M1 (AO1.1) A1 (AO1.1) M1 A1 [5]	Correct factorisation of denominator so correct form for partial fractions Using partial fractions, separating into two sums, re-indexing so that the summands have identical form and cancelling central terms.	Might see start and end terms explicitly. eg $\sum_{r=5}^n \frac{1}{r-1} - \sum_{r=8}^{n+3} \frac{1}{r-1}$ $= \left(\frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} \right) - \left(\frac{1}{n-1} + \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} \right)$ Might see formal substitution of index. eg Let $R = r + 3 \Rightarrow r + 2 = R - 1$ $\therefore \sum_{r=5}^n \frac{1}{r-1} - \sum_{r=5}^n \frac{1}{r+2}$ $= \sum_{r=5}^n \frac{1}{r-1} - \sum_{R=8}^{n+3} \frac{1}{R-1}$	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
			$= \frac{1}{4} + \frac{1}{5} + \frac{1}{6} - \frac{1}{n} - \frac{1}{n+1} - \frac{1}{n+2}$ $= \frac{37}{60} - \frac{1}{n} - \frac{1}{n+1} - \frac{1}{n+2}$		At least these terms	M1 can be ft from any A, B having opposite signs For M1, condone omission of $\frac{1}{7}$ or $-\frac{1}{n-1}$ Requires joined argument Must have either clear diagonal cancellations with one explicit cancellation or less clear cancellation with numerical and algebraic cancellation shown or described



Question			Answer/Indicative content	Marks	Part marks and guidance	
	b		$= \frac{37}{60}$ or awrt 0.617	B1 (AO2.2a) [1]	<u>Examiner's Comments</u> The mark scheme gives an algebraic demonstration of method of differences as the most efficient approach. Often, historically, teaching and textbooks have focused on the alternative approach on the mark scheme – namely one which writes down several numerical and algebraic terms from the sum and shows subsequent evidence of cancellation through crossing out. This alternative approach will continue to receive credit, but centres may wish to consider teaching the algebraic approach – certainly stronger candidates will enjoy the mathematical precision, and the completeness of the method will reduce any uncertainty as to just how many terms are required to convincingly demonstrate the alternative approach.	
			Total	6		

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Question		Answer/Indicative content	Marks	Part marks and guidance		
4	a	DR $u = x^2$ $3(\sqrt{u})^3 - 2(\sqrt{u})^2 - 5\sqrt{u} - 4 (=0)$ $3u\sqrt{u} - 5\sqrt{u} = 2u + 4 \Rightarrow u(3u - 5)^2 = (2u + 4)^2$ $u(9u^2 - 30u + 25) = 4u^2 + 16u + 16 \Rightarrow$ $9u^3 - 34u^2 + 9u - 16 = 0$ Alternative method DR $\alpha^2 \beta^2 \gamma^2 = (\alpha \beta \gamma)^2 = \left(-\frac{4}{3}\right)^2 = \frac{16}{9}$ $\alpha^2 \beta^2 + \beta^2 \gamma^2 + \gamma^2 \alpha^2$ $= (\alpha \beta + \alpha \gamma + \gamma \alpha)^2 - 2\alpha \beta \gamma (\alpha + \beta + \gamma)$ $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha \beta + \beta \gamma + \gamma \alpha)$ $u^3 - \left(\left(\frac{2}{3}\right)^2 - 2 \times \frac{-5}{3}\right)u^2 + \left(\left(\frac{-5}{3}\right)^2 - 2 \times \frac{4}{3} \times \frac{2}{3}\right)u - \frac{16}{9}$ $= u^3 - \frac{34}{9}u^2 + u - \frac{16}{9} = 0 \Rightarrow 9u^3 - 34u^2 + 9u - 16 = 0$	B1 (AO3.1 a) M1 (AO1.1) A1(AO3.2 a) B1 M1 M1 A1 [4]	Correct substitution chosen Oe Attempting to make substitution Rearranging and squaring to remove the square root(s) Rearranging to answer Writing the expression in terms of standard symmetrical forms	or preparation for substitution by removing odd powers. eg $x^2(3x^2 - 5)^2 = (2x^2 + 4)^2 \dots$...and then substituting $u(3u - 5)^2 = (2u + 4)^2$ Equation can be in x Must include one intermediate step N $\sum \alpha = \frac{2}{3}$, B $\sum \alpha \beta = \frac{-5}{3}$ $\alpha \beta \gamma = \frac{4}{3}$	Condone without factorisation of "2" N $\sum \alpha^2 = \frac{34}{9}$,



Question			Answer/Indicative content	Marks	Part marks and guidance		
					rearranging to answer	B $\sum \alpha^2 \beta^2 = 1$	
	b		DR $\frac{\sum \alpha^2 \beta^2}{\alpha \beta \gamma} = \frac{(\frac{9}{9})}{(\frac{4}{3})} \text{ or } \frac{1}{(\frac{4}{3})}$ $= \frac{3}{4}$	M1 (AO3.1a) 1(AO1.1) [2]	Their $\alpha^2 \beta^2 + \beta^2 \gamma^2 + \gamma^2 \alpha^2$ from part (a) $\pm \frac{4}{3}$ over	Strict ft	
					<u>Examiner's Comments</u> Far more candidates used a method involving specification point 4.05a (relationship between roots and coefficients) than 4.05b (substitution). Although either method is acceptable, the latter approach is probably more efficient in this case.		
			Total	6			



Question	Answer/Indicative content	Marks	Part marks and guidance
5	$\frac{4}{1-x^4} = \frac{A}{1-x} + \frac{B}{1+x} + \frac{Cx+D}{1+x^2}$ $\Rightarrow A(1+x)(1+x^2) + B(1-x)(1+x^2) + (Cx+D)(1-x^2) = 4$ $x=1: 4A=4 \Rightarrow A=1$ $x=-1: 4B=4 \Rightarrow B=1$ $x=0: A+B+D=4 \Rightarrow D=2$ $x^3: A-B-C=0 \Rightarrow C=0$ $I = \int_0^{\frac{1}{\sqrt{3}}} \left(\frac{1}{1-x} + \frac{1}{1+x} + \frac{2}{1+x^2} \right) dx$ $= \left[\ln \left(\frac{1+x}{1-x} \right) + 2 \tan^{-1} x \right]_0^{\frac{1}{\sqrt{3}}}$ $= \ln \left(\frac{\sqrt{3}+1}{\sqrt{3}-1} \right) + 2 \tan^{-1} \frac{1}{\sqrt{3}} \quad (-0) = \ln \left(\frac{\sqrt{3}+1}{\sqrt{3}-1} \right) + \frac{\pi}{3}$ $= \ln(2+\sqrt{3}) + \frac{\pi}{3} \text{ i.e. } a=2, b=3, c=3$ $\frac{4}{1-x^4} = \frac{A}{1-x} + \frac{B}{1+x} + \frac{D}{1+x^2}$ $\frac{4}{1-x^4} = \frac{A}{1-x^2} + \frac{D}{1+x^2}$ with $A=D=2$ by inspection Followe $\frac{2}{1+x^2} + \frac{1}{1-x} + \frac{1}{1+x}$ d by in integration section $\frac{4}{1-x^4} = \frac{A}{1-x^2} + \frac{D}{1+x^2}$ $\Rightarrow \int_0^{\frac{1}{\sqrt{3}}} \frac{4}{1-x^4} dx = \int_0^{\frac{1}{\sqrt{3}}} \left(\frac{2}{1-x^2} + \frac{2}{1+x^2} \right) dx$	M1 (AO3.1a) A1 (AO1.1) A1 (AO1.1) M1 (AO1.1) A1 (AO1.1) M0 M1 A1 A1 M1 A1 A1 [6]	Proper split to produce integrable integrand For one of the terms / constants For all terms / constants $\frac{1}{4}$ of these See below for other possibilities Correctly integration of their integrand without simplification - ignore limits Substitution - ignore - 0 Values must be stated But consider last 3 marks for correct integration Look to see the second split further on in question Look for the integration. If $\tanh^{-1}$ is used then give full marks here.



Question			Answer/Indicative content	Marks	Part marks and guidance
			$= \left[2 \tanh^{-1} x + 2 \tan^{-1} x \right]_{\frac{1}{\sqrt{3}}}^1 = \left(2 \tanh^{-1} \frac{1}{\sqrt{3}} + 2 \tan^{-1} \frac{1}{\sqrt{3}} \right)$ $= \ln \left(\frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} \right) + 2 \frac{\pi}{6} - 0$ $= \ln \left(\frac{\sqrt{3} + 1}{\sqrt{3} - 1} \right) + \frac{\pi}{3} = \ln \left(\frac{1 + 2\sqrt{3} + 3}{2} \right) + \frac{\pi}{3} = \ln \left(\frac{4 + 2\sqrt{3}}{2} \right) + \frac{\pi}{3}$ $= \ln(2 + \sqrt{3}) + \frac{\pi}{3}$		<u>Examiner's Comments</u> Many candidates appreciated that the integrand had to be split into partial fractions in order to perform the integration. The $\frac{D}{1+x^2}$ instead of $\frac{Cx+D}{1+x^2}$ inclusion of was not a correct split and so did not earn the first 3 marks. However, since $C = 0$ the result was the same and so the last 3 marks could be earned. The $\frac{4}{1-x^4} = \frac{2}{1+x^2} + \frac{2}{1-x^2}$ was only acceptable in two circumstances. (i) If $\frac{2}{1-x^2}$ was split further to give $\frac{2}{1-x^2} = \frac{1}{1+x} + \frac{1}{1-x}$ as a further process during integration. (ii) $\frac{2}{1-x^2}$ was seen to give $\tanh^{-1} x$ when integrated.
			Total	6	

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Question		Answer/Indicative content	Marks	Part marks and guidance	
	d	$\alpha = 1 + i\sqrt{2} = \sqrt{3} \left(\frac{1}{\sqrt{3}} + i \frac{\sqrt{2}}{\sqrt{3}} \right) = 3^{\frac{1}{2}} (\cos \theta + i \sin \theta)$ where $\cos \theta = \frac{1}{\sqrt{3}}$ $\sin \theta = \frac{\sqrt{2}}{\sqrt{3}} \Rightarrow \tan \theta = \sqrt{2}$ re $\beta = 1 - i\sqrt{2} = \sqrt{3} \left(\frac{1}{\sqrt{3}} - i \frac{\sqrt{2}}{\sqrt{3}} \right) = 3^{\frac{1}{2}} (\cos \theta - i \sin \theta)$ oe $\Rightarrow \alpha^n = 3^{\frac{n}{2}} (\cos n\theta + i \sin n\theta)$, $\beta^n = 3^{\frac{n}{2}} (\cos n\theta - i \sin n\theta)$ AG $\Rightarrow \alpha^n + \beta^n = 2 \times 3^{\frac{n}{2}} \times \cos n\theta$ $\Rightarrow \alpha^n + \beta^n + \gamma^n = 2 \times 3^{\frac{n}{2}} \times \cos n\theta + \frac{1}{2^n}$	M1 (AO2.1) A1 (AO1.1) M1 (AO2.1) A1 (AO2.1) [4]	Either α or β seen in mod / arg form For both of them - accept exponentials Derivation of α^n or β^n <u>Examiner's Comments</u> In part (b), rather than using the symmetry of roots, the working for part (c) was seen. So, some extra algebra was needed, creating more opportunities for errors. Part (d) proved to be a challenging question for most candidates. It was necessary to realise that the only way to find a simple answer for the nth power of each complex root was by a conversion to mod / arg form. Those that did realise this usually succeeded in obtaining the required response.	
		Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
7			Basis case: $1^3 + 2^3 + 3^3 = 36 = 4 \times 9$ Consider the sum $f(r) = r^3 + (r + 1)^3 + (r + 2)^3$ Assume that $f(r) = 9k$ for some $k \in \mathbb{Z}$ Then $f(r + 1) = f(r) + (r + 3)^3 - r^3$ $= f(r) + r^3 + 3r^2 \times 3 + 3r \times 9 + 27 - r^3$ $= f(r) + 9r^2 + 27r + 27$ $= 9k + 9(r^2 + 3r + 3) = 9k'$ for some $k' \in \mathbb{Z}$ So if true for r then true also for $r + 1$ But it is true for $r = 1$ so is true for all integers r	B1 (AO2.1) M1 (AO2.1) M1 (AO2.1) A1 (AO2.2a) A1 (AO2.5) [5]	Basis case - either 4×9 or "36 is divisible by 9" given explicitly. For getting started (Sight of $f(r)$ is not necessary) For finding $f(r + 1)$ Each term as a multiple of 9 Conclusion dependant on all other marks being earned <u>Examiner's Comments</u> Many candidates completed this question successfully. However, a significant number of candidates dropped marks with the formal statements required for an inductive proof, either in the initial base case or in the final conclusion.	
			Total	5		



Question		Answer/Indicative content	Marks	Part marks and guidance	
8	a	DR $\alpha + \beta = 2, \alpha\beta = 5$	B1(AO1.1) [1]	<u>Examiner's Comments</u> This should have been an easy first mark for candidates and indeed it was for the vast majority. It was pleasing to note that most candidates interpreted the command words 'Write down' correctly, meaning that no work should be needed; the values should have been seen in the quadratic equation given. A few candidates chose to find the actual complex values for α and β.  OCR support OCR's command words poster is available to download in both A2 and A4 versions from 'Assessment guides' at https://www.ocr.org.uk/qualifications/as-and-a-level/further-mathematics-a-h235-h245-from-2017/assessment/	



Question			Answer/Indicative content	Marks	Part marks and guidance		
	b		DR $\left(\alpha + \frac{1}{\beta}\right) + \left(\beta + \frac{1}{\alpha}\right) = \alpha + \beta + \left(\frac{\alpha + \beta}{\alpha\beta}\right) = 2 + \frac{2}{5} = \frac{12}{5}$ $\left(\alpha + \frac{1}{\beta}\right) \times \left(\beta + \frac{1}{\alpha}\right) = \alpha\beta + 2 + \frac{1}{\alpha\beta} = 7 + \frac{1}{5} = \frac{36}{5}$ $\Rightarrow 5x - 12x + 36 = 0 \text{ oe}$	M1(AO1.1a) A1(AO1.1) A1(AO2.2a) [3]	Attempt both sum and product of new roots in terms of original roots For one of $12/5$ or $36/5$ For both, correctly interpreted as quadratic.	DR so finding roots M0	
			Total	4			



Question			Answer/Indicative content	Marks	Part marks and guidance		
9			DR $\frac{x+1}{x^3-x^2+x-1} = \frac{x+1}{(x^2+1)(x-1)}$ $= \frac{Ax+B}{(x^2+1)} + \frac{C}{(x-1)}$ $\Rightarrow (Ax+B)(x-1) + C(x^2+1) = x+1$ $x=1: 2C=2 \Rightarrow C=1$ $x=0: -B+C=1 \Rightarrow B=0$ $x=2: 2A+5=3 \Rightarrow A=1$ $\Rightarrow \frac{x+1}{(x^2+1)(x-1)} = \frac{1}{(x-1)} - \frac{x}{(x^2+1)}$ $\Rightarrow \int \frac{x+1}{x^3-x^2+x-1} dx = \int \left(\frac{1}{(x-1)} - \frac{x}{(x^2+1)} \right) dx$ $\left[\ln(x-1) - \frac{1}{2} \ln(x^2+1) \right]_2^3 = \ln 2 - \frac{1}{2} \ln 10 + \frac{1}{2} \ln 5$ $= \frac{1}{2} \ln 2$	M1 (AO3.1a) M1 (AO2.1) M1 (AO3.1a) A1 (AO1.1) M1 (AO1.1) A1 (AO2.2a) [6]	Factorise denominator Split into partial fractions Valid method for finding A, B and C Substituting correct limits into their integral of the form $\alpha \ln(x-1) + \beta \ln(x^2+1)$ oe in correct form (not eg $\ln \sqrt{2}$)		
			Total	6			



Question			Answer/Indicative content	Marks	Part marks and guidance	
10		a	$\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a} \text{ used}$ $\left(x - \frac{b}{a}\right)\left(x - \frac{c}{a}\right) = 0$ $x^2 + \frac{b-c}{a}x - \frac{bc}{a^2} = 0$ $a^2 x^2 + a(b-c)x - bc = 0$	B1 (AO 1.1a) M1 (AO 3.1a) A1 (AO 1.1) A1 (AO 2.2a)	Or $\alpha' = \beta' = -\frac{b}{a} + \frac{c}{a}, \alpha' + \beta' = -\frac{b}{a} + \frac{c}{a}$ Correctly expanding brackets and collecting terms Or any non-zero integer multiple (ie correctly multiplying by a^2 (or ka^2 or a^3 etc) (could be done earlier))	
		b	Both having repeated roots $\Rightarrow \alpha = \beta$ and $a\beta = \alpha + \beta \Rightarrow \alpha^2 = 2\alpha$ so either $\alpha = \beta = 0 \Rightarrow b = c = 0$ or $\alpha = \beta = 2 \Rightarrow b < 0$ (or $a < 0$ & $c < 0$). But $a, b, c > 0$	M1 (AO 3.1a) E1 (AO 2.1) [2]	Assume repeated roots and set up equations. For establishing contradiction	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	$\Delta = (a(b - c))^2 - 4a^2(-bc)$ $\Delta = a^2(b^2 - 2bc + c^2 + 4bc)$ $= a^2(b + c)^2$ which is positive since a, b and c are real and both $a \neq 0$ and $b + c \neq 0$	M1 (AO 3.1a) A1 (AO 1.1) E1 (AO 2.4) [3]	Correct substitution into " $b^2 - 4ac$ " $a^2 > 0$ and $(b + c)^2 > 0$ alone is insufficient	
			Total	9		
11			DR Substi $y = \frac{1}{x}$ tute $\Rightarrow \frac{2}{y^3} - \frac{3}{y^2} + \frac{1}{y} + 4 = 0$ $\Rightarrow 4y^3 + y^2 - 3y + 2 = 0$ has $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ roots $\Rightarrow \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = -\frac{1}{4}$	M1 (AO 3.1a) A1 (AO 1.1) M1 (AO 1.1a) A1 (AO 1.1) [4]		
			Total	4		



Question			Answer/Indicative content	Marks	Part marks and guidance		
12		a	(i) All roots satisfy $ z^5 = a^5 \Rightarrow z = a$	E1(AO 2.2a) [1]	Clear justification that all points are equidistant from O		
		a	(ii) All points are distance a from O Each root of the form $a \operatorname{cis}\left(\frac{2k\pi}{5}\right)$ hence spaced at angular intervals of $\frac{2\pi}{5}$ around this Circle	B1(AO 1.1) E1(AO 2.4) [2]	Using the answer to part (ii)		
		a	(iii) $\arg(b \times e^{\frac{2\pi i}{5}}) = \arg(b) + \arg(e^{\frac{2\pi i}{5}})$ $= \frac{4}{5}\pi = \arg(c)$	E1(AO 2.2a) [1]	Multiplying any complex number by $e^{\frac{2i\pi}{5}}$ rotates it through an angle of $\frac{2\pi}{5}$, so $b \rightarrow$		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		a	(iv) $b^* = \operatorname{cis}\left(\frac{2\pi}{5}\right)^* = \operatorname{cis}\left(\frac{-2\pi}{5}\right) = \operatorname{cis}\left(\frac{8\pi}{5}\right) = e$	E1(AO 2.4) [1]	Or by noting that complex roots of a real polynomial occur in conjugate pairs (the pairing is obvious)	
		a	(v) a, b, c, d and e are the roots of $z^5 - a^5 = 0$ so sum of a, b, c, d and e is $-\frac{0}{1} = 0$	M1(AO 2.1) E1(AO 1.1) [2]	Accept any complete justification	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	All roots of equal magnitude $a \cos \frac{\pi}{5}$ All roots spaced at angles $\frac{2\pi}{5}$ of $\frac{2\pi}{5}$ r is negative real root $\Rightarrow z^5 + (a \cos \frac{\pi}{5})^5 = 0 \Rightarrow z^5 = 0$	M1(AO 3.1a) B1(AO 3.1a) A1(AO 2.2a) [3]	From any sensible right-angled triangle Must justify equally spread arguments Allow correct eqn. from 1st M1 only		
			Total	10			
13			$\alpha + \beta + \gamma = -2, \alpha\beta + \beta\gamma + \gamma\alpha = 1, \alpha\beta\gamma = -3$ $p = -(\alpha\beta + \beta\gamma + \gamma\alpha) = -1$ $q = \alpha\beta\gamma + \beta\gamma\gamma\alpha + \gamma\alpha\alpha\beta$ $= \alpha\beta\gamma(\alpha + \beta + \gamma) = 6$ $r = -\alpha\beta\gamma\gamma\alpha = -9$	B1(AO 1.1) B1(AO1.1) M1(AO2.1) A1(AO1.1) A1(AO1.1) [5]	soi Manipulation of roots		
			Total	5			



Question			Answer/Indicative content	Marks	Part marks and guidance	
14			$5x^2 + x + 12 \equiv A(x^2 + k) + (Bx + C)x$ $A = \frac{12}{k}, B = 5 - \frac{12}{k}$ Since A and B must be integers, k must divide 12. $B > 0$ gives $k = 3, 4, 6$ or 12	M1(AO1.1) M1(AO3.1a) A1(AO1.1) M1(AO3.1a) E1(AO2.2a) [5]	Attempt to find A, B and C in terms of k Attempt to express A or B in terms of k For both expressions Interpret the expression for A or B as an integer For this set only, having considered both A and B	Allow without $C = 1$ since it is independent of k
			Total	5		



Question			Answer/Indicative content	Marks	Part marks and guidance	
15		i	Either	M1	Substitute α into equation and rearrange	$\alpha^3 - k\alpha^2 - 2 = 0$
		i	$k = \alpha - \frac{2}{\alpha^2}$	A1	Obtain given answer a.e.f.	
		i	Or	M1	Substitute for k and x in terms of α and simplify	
		i		A1	Show simplification leads to consistency	e.g. "LHS = 0"
		i	Or	M1	Eliminate β and γ from symmetric functions	Don't penalise sign errors
		i		A1	Obtain given answer correctly	
					<u>Examiner's Comments</u>	
					The majority of candidates started by writing down some or all of the symmetric functions and then showed insufficient working to justify obtaining the given answer in part (i). Very few used the idea that a root satisfies the given equation and hence k can be found in terms of α .	
	ii			B1	State or use $(y) = u - iv$	
	ii			M1*	Use sum of roots = $(\pm)k$ (Can use $\Sigma a\beta$ with $a\beta$)	$\alpha + u + tv + u - tv = -(-k)$
	ii		$u = -\frac{2}{2\alpha^2}$ or better	DM1	Rearrange to get u	$\alpha + 2u = \alpha - \frac{2}{\alpha^2}$



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	Obtain correct answer <u>Examiner's Comments</u> Again, having obtained all 3 symmetric functions, many did not see the simplest way of finding the required result in parts (ii) and (iii), which often meant that algebraic errors were made. The incorrect result $(u + iv)(u - iv) = u^2 - v^2$ was seen very frequently.	
		iii		M1*	Use product of roots = $(\pm)2$ or $\sum \alpha\beta = 0$	In terms of u and v
		iii	$\alpha(u^2 + v^2) = 2$ or $2\alpha u + u^2 + v^2 = 0$	A1	Obtain correct answer	
		iii		DM1	Substitute for u and rearrange to get v^2	
		iii	$v^2 = \frac{2}{\alpha} - \frac{1}{\alpha^4}$	A1	Obtain correct answer a.e.f. <u>Examiner's Comments</u> Again, having obtained all 3 symmetric functions, many did not see the simplest way of finding the required result in parts (ii) and (iii), which often meant that algebraic errors were made. The incorrect result $(u + iv)(u - iv) = u^2 - v^2$ was seen very frequently.	
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
16		i	$4u^2 + 6u + k + 2 = 0$	M1	Substitute and attempt to simplify	
		i		A1	Obtain correct answer, must be an equation	
					Examiner's Comments	
					This part was done correctly by the majority of candidates. Too many omitted " $= 0$ " and so did not obtain an equation.	
		ii	Either			
		ii	$\frac{k+2}{4}$	M1	Use products of roots of new quadratic i.e. use $(\pm) c/a$	
		ii		A1ft	Obtain correct answer, from their quadratic	
		ii	Or			
		ii	$\frac{k+2}{4}$	M1	Use sum and product of roots of original equation	
		ii		A1	Obtain correct answer	
					Examiner's Comments	
					Most used the product of the roots of the derived equation, or used the sum and product of the roots of the given equation in an expanded expression. However a minority used the values from the derived equation in an expanded expression. A smaller minority found the roots of either the given or derived quadratic and attempted to use these to find the value of the given expression, usually without success.	
			Total	4		

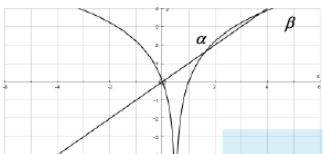


Question			Answer/Indicative content	Marks	Part marks and guidance	
17		i		M1	Attempt at complete expansion	
		i		A1	Obtain correct unsimplified answer	
		i		A1	Obtain given answer correctly	
					Examiner's Comments	
					Most derived the given answer correctly and showed sufficient working. Minor algebraic slips resulted in major loss of marks, but some showed little or no working to justify their answer.	
		ii	<i>Either</i>			
		ii	$\Sigma\alpha = -p, \Sigma\alpha\beta = -4, \alpha\beta\gamma = -3$	B1	State (anywhere) correct values for $\Sigma\alpha, \Sigma\alpha\beta, \Sigma\alpha\beta\gamma$	
		ii		M1	Express given expression as a single fraction	
		ii		A1	Obtain correct expression using (i)	
		ii		M1	Use their values for sum of roots etc. in their expression	
		ii	$\frac{16-6p}{9}$	A1	Obtain correct answer	
		ii	<i>Or</i>			
		ii		B1	Use substitution $1/\sqrt{u}$	
		ii	$9u^3 + (6p - 16)u^2 + (8 + p^2)u - 1 = 0$	M1	Rearrange appropriately and square out	
		ii	$\frac{16-6p}{9}$	A1	Obtain correct co-efficients of u^3 and u^2	
		ii		M1	Use $(+/-)b/a$ from their cubic	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	Obtain correct answer Examiner's Comments The expression in (i) was generally used to obtain a fraction in which the values of the symmetric functions could be substituted. Sign errors caused the major loss of marks. Some used the substitution $x = \frac{1}{\sqrt{u}}$ and then rearranged correctly in order to square out and obtain a correct cubic equation. The correct value of the required expression can then be easily found. Some used a substitution $x = \frac{1}{u}$ and then a similar approach to (i) to find the value of the required expression. These last two methods usually resulted in more lengthy solutions.	
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Question			Answer/Indicative content	Marks	Part marks and guidance	
18		i	$x = 0$ in equation satisfies as $e^0 = 1$.	B1		Examiner's Comments There were many good answers to this question; candidates usually find the questions on this topic straightforward. It was rare to find a script in which this part did not yield the one mark.
		ii	(a) 	B1	Asymptote between $x = 0$ and where it crosses x axis.	Allow one branch.
		ii		B1	+ve roots clear	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii			LH branch going through origin LH branch does not have to be complete	SC1 $y = (\ln(2x-1))^2$ Examiner's Comments There were many good answers to this question; candidates usually find the questions on this topic straightforward. As with the other questions, the drawing of the graph was not good. Many failed to appreciate the fact that $x = \frac{1}{2}$ was an asymptote and so struggled to draw what they thought was the correct graph that fulfilled the result of (i). A significant number of candidates drew the section where the curve approached the asymptote as a "spike" (from graphical calculators without thinking about the nature of the curve?). Candidates who rewrote the expression as $y = 2\ln(2x - 1)$ lost the part of the graph for $x = \frac{1}{2}$. The best candidates marked the asymptote using a dashed line.



Question			Answer/Indicative content	Marks	Part marks and guidance	
	ii		(b) Staircase seen near middle root to be converging to β .	B1	Either starting point shown with vertical line from axis to curve or arrows on staircase lines	Follow through their curve where there are two positive roots Examiner's Comments There were many good answers to this question; candidates usually find the questions on this topic straightforward. The 'staircase' needs to move vertically to the curve and horizontally to the line and there needed to be some indication of the start (labelling x_1 or a vertical line starting on the x -axis or arrows).
	ii		(c) $x_1 = 3.75$ $x_2 = 3.743604...$ Leading to 3.733	B3	For correct answer B2 for 3.734 B1 for sight of 3.7436...	Examiner's Comments There were many good answers to this question; candidates usually find the questions on this topic straightforward. The fixed point iteration was very slow and it needed at least 11 iterations to achieve accuracy to 3 decimal places. Most candidates did quote their answers to 3 decimal places but did not achieve the accuracy in the 3 rd decimal place that was required.
	iii		$f(x) = (2x-1)^2 - e^x$	B1	$f'(x)$ correct soi by x_2	f(x) correct and their $f'(x)$
	iii		$\Rightarrow f'(x) = 4(2x-1) - e^x$	M1	Use of formula soi by x_2	
	iii		$\Rightarrow x_{r+1} = x_r - \frac{(2x-1)^2 - e^x}{4(2x-1) - e^x}$	A1	x_2 to 2dp or better	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\Rightarrow x_2 = 1.629382\dots, x_3 = 1.629053$ Root = 1.6291	A1	Correct root stated to 5sf	At least 2 iterates shown Examiner's Comments There were many good answers to this question; candidates usually find the questions on this topic straightforward. Newton-Raphson has a much faster rate of convergence and β to 5 significant figures was achieved very quickly. Candidates should check the accuracy required in the question; some gave their answer to 5 decimal places instead of 5 significant figures. It should be noted that the correct Newton-Raphson formula with the correct $f'(x)$ was required. Some candidates rewrote the equation in a different way, achieving an $x = g(x)$ iterative formula that converged to β. However, this is not the Newton-Raphson formula that was required here and so merely finding β did not earn any credit. Likewise an incorrect $f'(x)$ will result in an iterative formula that converged to β, but again this was not the Newton-Raphson method and so earned no credit. Both of these erroneous processes to find β demonstrated a very much slower rate of convergence – the initial value $x = 1.6$ is already accurate to 1 decimal place and so accuracy to at least 4 decimal places is achieved in only 3 iterations

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Question			Answer/Indicative content	Marks	Part marks and guidance	
						using the Newton-Raphson formula.
			Total	12		
19			$\frac{A}{x-1} + \frac{Bx+C}{x^2+4}$ $\Rightarrow 5x \equiv A(x^2+4) + (Bx+C)(x-1) [+D(x-1)(x^2+4)]$ Equate coefficients or substitute values for x $\Rightarrow A = 1$ $B = -1$ $C = 4$ $\Rightarrow \frac{5x}{(x-1)(x^2+4)} = \frac{1}{(x-1)} + \frac{4-x}{(x^2+4)}$	B1 M1 A1 A1 A1	Sight of expression For Equating 3 coeffs or sub 3 times For one value (not D) For 2nd and 3rd values (not D) For final answer expressed properly <u>Examiner's Comments</u> Most scored well on this first question. A few wrote as their partial fractions $A + \frac{B}{x-1} + \frac{Cx+D}{x^2+4}$ For those that obtained A = 0 all that was lost was a little time but it did add an extra opportunity for error.	Allow addition of constant
			Total	5		



Question			Answer/Indicative content	Marks	Part marks and guidance	
20		i	$\frac{x^2 - 25}{(x-1)(x+2)} = A + \frac{B}{(x-1)} + \frac{C}{(x+2)}$ $x^2 - 25 = A(x-1)(x+2) + B(x+2) + C(x-1)$ 3 processes of equating coefficients or substituting: e.g. $x = 1 \Rightarrow -24 = 3B \Rightarrow B = -8$ $x = -2 \Rightarrow -21 = -3C \Rightarrow C = 7$	M1	Splitting in correct way to give partial fractions (may be seen anywhere)	
		i	coeff of x^2 : $A = 1$	B1		
		i	$\frac{x^2 - 25}{(x-1)(x+2)} = 1 - \frac{8}{(x-1)} + \frac{7}{(x+2)}$	A1		

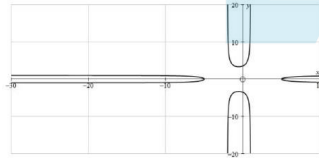
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Question			Answer/Indicative content	Marks	Part marks and guidance	
		i		A1	For C Examiner's Comments Some candidates did not recognise that the fraction was improper and started by using the form $\frac{A}{(x-1)} + \frac{B}{(x+2)}$ instead of $\frac{A}{(x-1)} + \frac{B}{(x+2)} + C.$ Many divided out to get $1 - \frac{x+23}{(x-1)(x+2)}$ but then lost the minus sign. Candidates who tried using the form $\frac{A}{(x-1)} + \frac{Bx+C}{(x+2)}$ were only credited with the method mark if they subsequently split the second fraction; most did not.	
		ii	$x = 1, x = -2$	B1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$y = 1$	B1	Examiner's Comments Most candidates knew that the vertical asymptotes were at $x = 1$ and $x = -2$. Many also knew that there was a horizontal asymptote at $y = 1$ although some gave $y = 0$ and some tried for an oblique asymptote. However, 90% of candidates obtained full marks in this part.	
		iii	$y = 1 \Rightarrow (x - 1)(x + 2) = x^2 - 25$	M1	Examiner's Comments Again, a significant majority achieved full marks. Those who did not usually equated the function to 0 rather than 1 in spite of getting (ii) correct.	
		iii	$x^2 + x - 2 = x^2 - 25 \Rightarrow x = -23$	A1		
		iv		B1	4 bits as shown, roughly symmetric about axes, approaching asymptotes	Ignore any graph of $y = f(x)$
		iv		B1	Lh side crosses asymptotes and upper section approaches from above and lower section approaches from below Examiner's Comments A few excellent graphs were seen, many graphs showing the four parts fairly well. Only a minority showed the part on the left-hand side crossing the horizontal lines $y = \pm 1$ and then approaching $y = 1$ from above and $y = -1$ from below.	
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
21		i	$\alpha + \beta = -\frac{1}{k}, \alpha\beta = 1$	B1	State correct values	
					Examiner's Comments	
					This was generally answered correctly, sign slips or failing to include the coefficient of x^2 being the most common errors.	
		ii	<i>Either</i>	M1	Expand expression	
		ii	$\alpha\beta + \frac{1}{\alpha\beta} + \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$	M1	Use correct process for $\alpha^2 + \beta^2$	
		ii		A1	Obtain correct expression	
		ii	<i>Or</i>	M2	State $\alpha = \frac{1}{\beta}$ and substitute into and given expression	
		ii	$(\alpha + \beta)(\beta + \alpha)$	A1	Obtain correct expression	
		ii		M1	Use their value(s) in their expression	
		ii	$\frac{1}{k^2}$ or k^{-2}	A1	Obtain correct single term answer	
					Examiner's Comments	
					Expansion of the given expression and using the results from (i) was the most popular method, with most being able to deal with $\alpha^2 + \beta^2$ in a sensible way. A significant minority saw that $\frac{1}{\alpha} = \beta$ and $\frac{1}{\beta} = \alpha$ and so could find the required value in a very neat way.	
			Total	6		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
22		i		B1	Use given substitution correctly in LHS of equation	
		i		M1	Rearrange and square to eliminate $\sqrt{u}$ or multiply by $u^{\frac{3}{2}}$ $u^{\frac{3}{2}} + 4u^{\frac{1}{2}} - 3$	
		i	$u^3 + 8u^2 + 16u - 9 = 0$	A1	Obtain correct answer, must be an equation = 0 Examiner's Comments Most candidates substituted correctly, but then did not know how to rearrange their equation before squaring it to obtain the new cubic equation. Many simply squared each term of their new equation.	
		ii	Either $\alpha\beta\gamma = -3$	B1	State or use correct result	
		ii	$\Sigma\alpha^2 = -8$ $\Sigma\alpha^2\beta^2 = 16$	B1B1	Use correct result, using correct (i) or using an identity involving $\Sigma\alpha = 0$, $\Sigma\alpha\beta = 4$	
		ii		M1*	Obtain an identity connecting $\Sigma\alpha^4$ and $(\Sigma\alpha^2)^2$	
		ii	$(\Sigma\alpha^2)^2 = \Sigma\alpha^4 + 2\Sigma\alpha^2\beta^2$	A1	Obtain a correct answer	
		ii		DM1	Use their values in their expression	
		ii	29	A1	Obtain correct answer, c.w.o.	
		ii	Or $\alpha\beta\gamma = -3$	B1	State or use correct result	
		ii	$\Sigma\alpha = 0$, $\Sigma\alpha\beta = 4$, $\Sigma\alpha^2 = -8$,	B1B1	Use any 2 correct B1, other 2 correct B1	
		ii	$\Sigma\alpha^2\beta^2 = 16$	M1	Expand $(\alpha + \beta + \gamma)^4$ and get expression involving symmetric functions only	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\Sigma \alpha^4 + 4 \Sigma \alpha^2 \cdot \Sigma \alpha\beta + 6 \Sigma \alpha^2\beta^2 + 8\alpha\beta\gamma \Sigma \alpha$	A1	Obtain correct expression	
		ii		M1	Use their values in their expression	
		ii		A1	Obtain correct answer, c.w.o.	
					Examiner's Comments	
					Those candidates who used the symmetric functions of the new cubic equation usually made good progress as the required identity is fairly easy to establish. Those who tried to use the symmetric functions of the original equation were usually unable to derive a correct identity and just produced many lines of incorrect working.	
			Total	10		



Question		Answer/Indicative content	Marks	Part marks and guidance	
23		Either $\sum \alpha = -\frac{6}{k}, \sum \alpha\beta = \frac{1}{k}$ $\sum \alpha\beta + 2\sum \alpha + 3$ $3 - \frac{11}{k}$ Or $ku^3 + (6 - 3k)u^2 + (3k - 11)u + 2 - k = 0$ $3 - \frac{11}{k}$	B1B1 M1 A1 M1 A1 B1 M1 A1A1 M1 A1	Correct values stated or used Expand brackets Obtain correct expression a.e.f. Use their values, in terms of k, for $\sum \alpha$ and $\sum \alpha\beta$ Obtain correct answer a.e.f. State or use substitution $x = u - 1$ Expand and attempt to simplify coefficients Obtain at least correct 1st and 3rd terms Use their "$\frac{c}{a}$" Obtain correct answer a.e.f. Examiner's Comments In using the symmetric functions for the given cubic, many candidates made sign errors. Those candidates who used the correct substitution usually answered the question totally correctly. However the incorrect substitution, $x = u + 1$, was seen on quite a few occasions. Some candidates used a mixture of the two methods, but usually managed to score some of the marks.	



Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	6	



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Question			Answer/Indicative content	Marks	Part marks and guidance	
24		i	$2u^3 + 12u^2 + 27u + 25 = 0$	M1	Substitute and attempt to simplify	Missing = 0 is an error
		i		A2	Obtain correct equation, A1 for only 1 error <u>Examiner's Comments</u> A surprisingly large number of candidates made errors in expanding $(u + 2)^3$ or in simplifying a correct expression into a cubic equation with correct coefficients, often omitting the “= 0”.	
		ii	Either	M1	Combine 3 terms with correct denominator	Must be $\pm c/a$ and $\pm d/a$ for M1 ft for their answer in (i)
		ii	$\frac{\sum \alpha'\beta'}{\alpha'\beta'\gamma'}$	A1	Obtain correct expression in their notation	
		ii		M1	Attempt to use values from (i) correctly	
		ii	$-\frac{27}{25}$	A1ft	Obtain correct answer with no errors seen	
		ii	Or	M1	Use substitution $y = \frac{1}{u}$	
		ii	$25y^3 + 27y^2 + 12y + 2 = 0$	A1ft	Obtain correct cubic equation, from their (i)	
		ii		M1	Use correct symmetric function	
		ii		A1ft	Obtain correct answer	
		ii	Or	M1	Combine 3 terms with correct denominator	
		ii	$\frac{\sum \alpha'\beta'}{\alpha'\beta'\gamma'}$	A1	Obtain correct expression in their notation	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		M1	Expand numerator and denominator and use values from original equation correctly	Condone $\pm$, but must be “/2”
		ii		A1	Obtain correct answer with no errors seen	
					<u>Examiner's Comments</u> This part proved to be quite testing. Many simply wrote down the value of “ $-b/a$ ”, or its reciprocal, from the new cubic equation. Others who combined the three fractions correctly, then expanded numerator and denominator but used the values of the symmetric functions from the new cubic equation, rather than the original equation. However, good candidates saw that a second substitution, $y = \frac{1}{u}$, could be used to find the required value.	
			Total	7		

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